COMPUTER NETWORKS

756341

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Even Semester 2024-25
B. Tech. (CSE/AIML&IIoT/BCT/CCV/DA/CSF), 2 Year, IV Semester
BCSC 0008 & Computer Networks

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all questions
2. No B sheet will be given

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A large university has multiple departments, each with its own star topology, and the departments are connected in a ring topology. What challenges might arise in above given hybrid topology? Also, explain how one can ensure smooth communication between departments in the given scenario.	3	2	U	P
2	A 4000 km long trunk operates at 1.623 Mbps and it is used to transmit 32 Bytes frames. If it uses sliding window protocol, calculate the Transmission delay and Propagation delay in (microsec). Assume propagation speed of 5 microsec/km.	3	3	A	C
3	How NRZ-L Line Coding Scheme differs with NRZ-I Line Coding Scheme. Draw the Manchester and Differential Manchester coding for the bit stream 100110101100.	3	2	A	C

4	Which transmission medium type is employed for wide-area communication? List the main applications of both guided and unguided media. Identify the major categories of guided media.	3	2	R	P
5	Assume that Source (S) and Destination (D) are connected through an intermediate Router. How many times a packet has to visit the network layer, data link layer, and physical layer during a transmission from S to D. Draw the complete diagram of this process from Source S to Destination D.	3	1	U	F

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain Cyclic Redundancy Check (CRC). Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X^3+X+1. Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. What is the value of the check bit sequence $c_2c_1c_0$?	5	4	An	C
7	Define Transmission impairments. Describe the various causes why signals can be impaired during transmission. Suppose a signal travels through a transmission medium and its power is reduced to one fourth during transmission. Calculate the attenuation (loss of power) in this case.	5	2	A	C
8	Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following: The time taken for processing the data frame by the receiver is negligible. The time taken for processing the acknowledgement frame by the sender is negligible.	5	4	An	M

	The sender has infinite number of frames available for transmission. The size of the data frame is 2,000 bits and the size of the acknowledgement frame is 10 bits. The link data rate in each direction is 1 Mbps (= 10^6 bits per second). One-way propagation delay of the link is 100 milliseconds. What is the minimum value of the sender's window size required in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50%.				